

GEAR STORM 2.0

Official Rules & Regulations

1. General Rules

1.1 Number of Teams

The competition will accommodate a maximum of **30 teams**, comprising:

- **15 teams in the Beginner Category**
- **15 teams in the Advanced Category**

1.2 Competition Format

The competition will consist of **two rounds**:

- **Round 1 (R1)**
- **Final Round**

The **top 6 teams from each category** in Round 1 will qualify for the Final Round.

1.3 Category Merge Rule

If either the Beginner or Advanced Category has **fewer than 10 registered teams**, the two categories will be merged into a single competition.

To account for the different constraints of the two categories, teams will be ranked using **Normalized Time**:

Normalized Time = Team's Adjusted Time ÷ Average Adjusted Time of all finishing teams in the team's original category for that round

Where:

Adjusted Time = Recorded Completion Time + Applicable Penalties

- Only teams that successfully complete the course (**non-DNF teams**) will be considered when calculating the category average.

- DNF teams will be ranked below all teams with a valid Normalized Time.
- If a category has no finishing team, the organizers will determine and announce an appropriate ranking method before the results are finalized.

1.4 Team Size

Each team must consist of a minimum of **2 members** and a maximum of **4 members**.

All members must belong to the **same college/institution**. Teams may be interdisciplinary, with members from different branches, departments, or academic years.

1.5 Participant Identification

All participants must carry a **valid college/event ID card** throughout the event.

1.6 No Test Runs

No test or practice runs will be permitted on the official arena before the competition, unless explicitly authorized by the organizers.

1.7 Reporting

All teams must report at the designated time for:

- Registration
- Robot inspection
- Technical briefing
- Their scheduled race

Teams must follow the announced event schedule. Late arrival may result in the team being denied participation at the discretion of the organizers.

1.8 Judges' Decision

The decisions of the judges regarding **penalties, rule violations, qualification, disqualification, safety, and race results shall be final and binding**.

1.9 Ranking

The winner will be the team with the **lowest Adjusted Time**, after all applicable penalties have been added.

1.10 Common Arena

Both Beginner and Advanced teams will compete on the **same official track and arena**.

1.11 Prize Pool

Beginner Category:

- 1st Place — ₹5,000
- 2nd Place — ₹3,000
- 3rd Place — ₹2,000

Advanced Category:

- 1st Place — ₹10,000
 - 2nd Place — ₹7,000
 - 3rd Place — ₹3,000
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2. Robot Specifications

2.1 Dimensions

The robot must fit within **25 cm × 25 cm × 25 cm (L × B × H)**.

A tolerance of **±5% on each dimension** will be permitted.

All components attached to the robot, including antennas and other protruding parts, will be considered during dimensional inspection.

The robot must remain within the permitted dimensions throughout the race. Deployable or extendable mechanisms that cause the robot to exceed the permitted dimensions are not allowed.

2.2 Weight

The robot must not exceed **3 kg**.

A tolerance of **+5%** will be permitted, making **3.15 kg the maximum acceptable weight**.

The measured weight includes the battery and all components mounted on the robot.

2.3 Battery

Only the following battery configurations are permitted:

- **3S Li-Po:** 11.1 V nominal, 12.6 V maximum fully charged

- **3S Li-ion:** 11.1 V nominal, 12.6 V maximum fully charged

Any other battery type or configuration requires **prior approval from the organizers**.

2.4 Battery Safety

- Only **one battery pack** may be used to power the robot during a run.
- Connecting multiple battery packs in series or parallel is not permitted unless approved by the organizers.
- The battery must be securely mounted on the robot.
- Battery terminals must be adequately insulated to prevent accidental short circuits.
- Damaged, swollen, leaking, exposed, or otherwise unsafe batteries will not be permitted.
- Li-ion battery packs must have appropriate protection/BMS.
- Batteries must be charged using a suitable charger and handled according to the manufacturer's safety recommendations.
- Battery charging will only be permitted in areas designated by the organizers.

2.5 Wired Robots

Wired robots are permitted.

The control cable must remain **slack throughout the entire run**.

Pulling, pushing, lifting, or otherwise manipulating the cable to assist the robot's movement is strictly prohibited and may result in **disqualification (DQ)**.

2.6 Wired or Wireless Operation

Robots may be either **wired or wireless**, subject to the control restrictions applicable to their category.

2.7 Robot Construction

Complete ready-made robots, RC cars, robot kits, or pre-assembled robotic platforms capable of directly performing the competition task are **not permitted**.

Individual commercially available components such as motors, wheels, sensors, microcontrollers, batteries, and motor drivers may be used.

The **chassis must be designed and built by the participating team**. A commercially pre-made robot chassis may not be used.

2.8 Track Safety

Robots must not contain any component, mechanism, or material that may **damage, scratch, cut, deform, or otherwise interfere with the track or arena.**

Sharp edges, dangerous projections, exposed hazardous parts, abrasive attachments, or hazardous materials are prohibited.

Any damage(s) to the track will lead to **disqualification.**

2.9 Mechanical Integrity

The robot must remain mechanically intact throughout the run.

If any component or part of the robot becomes detached and remains on the arena during the run, the team will be **disqualified.**

2.10 Post-Inspection Modifications

Once a robot has passed inspection, no major modification or replacement affecting its **dimensions, weight, battery, control system, category eligibility, or performance** may be made without approval from the organizers.

3. Arena Specifications

3.1 Arena Dimensions

The exact dimensions and final layout of the arena will be **disclosed on the day of the event.**

3.2 Track Conditions

The track surface and course line may contain **minor unevenness, irregularities, or non-uniform sections.**

Teams are expected to design their robots to handle such conditions.

3.3 Inclined Sections

The course may contain sudden changes in gradient, including inclined and declined sections.

The inclination of any such section will **not exceed 30° from the horizontal.**

3.4 Obstacles

The arena may contain obstacles such as:

- Switch Gate
- Speed Breakers
- Marble Pit
- Rotating Disk
- See-saw
- Ramps
- Other physical or mechanical challenges

The final combination and placement of obstacles will be disclosed on the day of the event.

3.5 Arena Configuration

The organizers reserve the right to make reasonable modifications to the arena or obstacle arrangement before the competition for **safety, technical, or logistical reasons**.

The final configuration announced by the organizers will be considered official.

4. Competition & Race Rules

4.1 Official Attempts

Each team will receive **one official attempt in Round 1**.

Teams qualifying for the Finals will receive **one official attempt in the Final Round**.

4.2 Starting Procedure

The robot must be positioned completely within the designated starting zone.

The run will begin only after the official start signal.

The robot must start under its own power and may not be pushed or pulled to gain an advantage.

4.3 Finish Procedure

The timer will stop when the robot **completely crosses the designated finish line under its own power**.

Physically pushing or pulling the robot across the finish line will not count as completion.

4.4 Checkpoints

Clearly marked checkpoints will be placed at intervals along the track.

A checkpoint will be considered successfully crossed when the **robot completely crosses the designated checkpoint marker/line**.

4.5 Restart After Penalty

Whenever a penalty requires the robot to be repositioned, it must restart from the **most recently crossed checkpoint**.

The robot may not be placed beyond that checkpoint.

4.6 Touch Penalty

Touching the robot during the run is permitted but will incur an increasing time penalty.

Touch	Penalty
1st	+5 sec
2nd	+8 sec
3rd	+11 sec
4th	+14 sec
Each subsequent touch	+3 sec over the previous penalty

Only the designated **Pilot and Pit/Support Member** may touch the robot.

After a team touch, the robot must be repositioned at the most recently crossed checkpoint before continuing, unless otherwise instructed by the judges.

Contact initiated by an event official for safety or operational reasons will **not count as a team touch**.

4.8 Obstacle Skipping

A team may skip an obstacle, but a time penalty will be imposed.

An obstacle is considered skipped when the robot **fails to negotiate the designated obstacle and proceeds beyond it without completing the required obstacle section.**

Skipped Obstacle	Penalty
1st	+15 sec
2nd	+30 sec
3rd	+45 sec
4th	DNF

A maximum of **3 obstacles** may be skipped during a run.

Teams may not manually move or place the robot beyond an obstacle solely to avoid completing it.

4.9 DNF

DNF (Did Not Finish) may be recorded if:

- The robot fails to complete the course within the maximum allotted time.
- The team skips more than 3 obstacles.
- The robot becomes incapable of continuing.
- The team fails to satisfy the conditions required for a valid finish.
- Any other condition specified by the judges results in termination of the run.

4.10 Repairs During a Run

Repairs or replacement of components during an official run are **not permitted**.

If the robot becomes incapable of continuing due to a technical or mechanical failure, the run will normally be terminated.

4.11 Event-Side Failure

A re-run may be authorized only if the judges determine that the interruption or failure was caused by:

- Arena malfunction
- Event equipment failure

- Event personnel
- Confirmed interference caused by the event
- Another confirmed event-side issue

A team's own mechanical, electrical, software, battery, or wireless-control failure will **normally not qualify for a re-run**.

4.12 External Assistance

No person other than authorized team members and event officials may physically assist, control, push, pull, reposition, repair, or otherwise influence the robot during a run.

4.13 Interference

Deliberately obstructing, touching, interfering with, or damaging another team's robot or the arena is strictly prohibited and may result in **immediate disqualification**.

4.14 Wireless Interference

Teams using wireless control are responsible for the reliable operation of their communication system.

Deliberate interference with another team's wireless communication is prohibited and may result in disqualification.

5. Team Members Near the Track

5.1 Authorized Personnel

Only **two members per team** may be present in the designated track-side area during a run:

1. **Pilot** — responsible for controlling the robot where applicable.
2. **Pit/Support Member** — responsible for permitted assistance.

5.2 Other Team Members

Any remaining team members must remain in the designated team/spectator area and will not be permitted near the track during the run.

5.3 Pit Assistance

The Pit/Support Member may not operate the robot or provide physical assistance to the robot unless specifically instructed by an event official.

6. Category-Wise Rules

6.1 Beginner Category

6.1.1 Budget Limit

The total reference value of the robot must be **₹5,000 or less**.

A tolerance of up to **₹500** may be permitted for minor pricing discrepancies during verification.

A robot valued above **₹5,500** will not be eligible for the Beginner Category.

6.1.2 Bill of Materials (BOM)

Every Beginner team must submit a complete **Bill of Materials (BOM)** containing:

- Component name
- Quantity
- Reference price
- Total component cost

The BOM will be verified by the organizers during inspection.

6.1.3 Component Valuation

All components will be valued according to their **current/reference market price**, irrespective of whether they were:

- Purchased recently
- Previously owned
- Borrowed
- Donated
- Obtained free of cost

For components available on **Robu.in**, the current listed price on Robu.in will be used as the primary reference.

For components not available on Robu.in, the organizers may use another recognized Indian electronics/robotics supplier as a reference.

6.1.4 Components Included in Budget

The valuation may include:

- Chassis material
- Motors
- Wheels
- Motor drivers
- Microcontrollers
- Sensors
- Batteries
- PCBs
- 3D-printed parts
- Wires
- Connectors
- Switches
- Fasteners
- Other components physically used in the robot

Shipping charges, discounts, coupons, or promotional offers will not normally be deducted from the reference value.

6.1.5 Dedicated RF/RC Restriction

To maintain a fair and beginner-friendly competition, the use of **dedicated external RF transmitter/receiver systems or commercial remote-control systems for manual driving is prohibited.**

Examples include:

- 433 MHz RF remote-control modules
- nRF24L01-based remote-control systems
- HC-12-based systems
- FlySky or similar RC transmitter/receiver systems
- Similar dedicated wireless remote-control systems

6.1.6 Permitted Electronics

Standard microcontrollers and development boards such as **Arduino, ESP32, ESP8266, STM32**, etc. may be used for onboard sensing, processing, and autonomous control.

Such devices may not be used to circumvent the prohibition on manual remote control.

6.2 Advanced Category

6.2.1 Budget

The total reference value of the robot must be **above ₹5,500**.

There is **no upper budget limit**.

6.2.2 Component Freedom

Advanced teams may use commercially available:

- Motors
- Sensors
- Controllers
- Motor drivers
- Communication systems
- Mechanical components
- Other electronic components

subject to the general robot safety, dimension, weight, battery, and arena requirements.

6.2.3 Control Freedom

Manual, wireless, autonomous, and hybrid control methods are permitted.

6.2.4 Wireless Systems

Dedicated RF/RC systems, Bluetooth, Wi-Fi, and other suitable wireless communication methods are permitted, provided they do not deliberately interfere with other teams or violate safety requirements.

7. Inspection, Safety & Compliance

7.1 Mandatory Robot Inspection

Every robot must pass an inspection before participating.

The organizers may verify:

- Dimensions
- Weight
- Battery
- Construction
- Chassis
- Mechanical safety
- Category eligibility
- BOM
- Control system

7.2 Inspection Failure

A robot that fails inspection may be denied participation until the violation is corrected and the robot passes inspection.

Corrections must be completed within the time permitted by the organizers.

7.3 Category Compliance

A team may not modify its robot after inspection in a manner that changes its declared category or provides an unfair advantage.

7.4 Emergency Authority

Judges or event officials may immediately stop or switch off a robot if it is considered:

- Unsafe
- Malfunctioning
- Capable of damaging the arena
- A risk to participants or equipment

7.5 Battery Charging

Battery charging is permitted only in areas designated by the organizers.

Damaged or unsafe batteries may not be charged or used at the venue.

7.6 Electrical Safety

Exposed conductors, unsafe wiring, short-circuit risks, or other electrical conditions considered hazardous by the organizers may result in rejection during inspection.

7.7 Disqualification

A team may be disqualified for:

- Using prohibited equipment
 - Violating category restrictions
 - Deliberately damaging the arena
 - Manipulating a control cable to gain an advantage
 - Receiving unauthorized assistance
 - Interfering with another team
 - Leaving components on the track
 - Cheating
 - Misconduct
 - Violating any other official rule
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8. Timing, Penalties & Tie-Breaking

8.1 Adjusted Time

Adjusted Time = Recorded Completion Time + All Applicable Time Penalties

The team with the lowest Adjusted Time will receive the higher ranking.

8.2 Penalty Recording

All penalties incurred during a run will be recorded by the judges and added to the team's completion time.

8.3 Tie-Breaker

If two or more teams have the same Adjusted Time:

1. The team with **fewer total penalties** will be ranked higher.
2. If the tie remains, the organizers may conduct a tie-breaker or use available race data to determine the ranking.

8.4 Normalized Time for a Merged Category

If the categories are merged under Rule 1.3, Normalized Time will be calculated using each team's **Adjusted Time**, ensuring that all applicable penalties are included before normalization.

9. Protests, Conduct & Organizer Authority

9.1 Clarification/Protest

A team wishing to raise a concern regarding its run or a penalty must approach the designated event official **within 5 minutes of completing the run**.

Concerns raised after this period may not be considered.

9.2 Professional Conduct

All participants are expected to maintain professional and respectful conduct toward:

- Other teams
- Judges
- Volunteers
- Organizers
- Event staff

Abusive behavior, deliberate disruption, cheating, or misconduct may result in disqualification.

9.3 Rule Interpretation

If a situation arises that is not explicitly covered by these rules, the judges and organizers will interpret the rules in the interest of **safety, fairness, and the smooth conduct of the competition**.

Their decision will be final and binding.

9.4 Organizer Authority

The organizers reserve the right to modify the schedule, arena configuration, inspection procedure, or operational details when necessary for **safety, technical, or logistical reasons**.

Any such changes will be communicated to participating teams.